FARAAZ MOHAMMED

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Full Australian working rights (Temporary Graduate visa, subclass 485) — no sponsorship required • Available immediately

PROFILE

Master of Artificial Intelligence graduate (RMIT, 2026) building generative-AI and machine-learning solutions in Python: a RAG chatbot, an MCP-based agent delegation tool with a fine-tuning loop, an LLM model router for cost, and data pipelines over 7.2M records for an industry partner. Turns plain-language requests into requirements and decision logs, and has deployed a Python service to the cloud with CloudFormation. Co-author of a peer-reviewed computer-vision paper. Seeking the AI Engineer role at One51; open to relocating to Sydney.

TECHNICAL SKILLS

Generative AI: LLMs, prompt engineering, retrieval-augmented generation (RAG), AI orchestration and agents (MCP servers, tool delegation), model routing for cost, Anthropic API (prompt caching), LLM fine-tuning (Unsloth, QLoRA, GGUF), Ollama

Machine learning & evaluation: scikit-learn, XGBoost, OpenCV; evaluation design (precision/recall, IoU, sensitivity analysis, ground-truth construction)

Data engineering & analytics: Python (primary), SQL, pandas, PostgreSQL (SQLAlchemy), Kestra pipelines, GeoPandas, Parquet / GeoPackage / GeoJSON

Cloud, APIs & delivery: AWS (CloudFormation, ECS Fargate, RDS, CloudFront, SNS), FastAPI, REST/JSON, Docker, Git / GitHub, GitHub Actions CI, Vercel

Consulting: requirements and acceptance criteria, decision logs, cross-team handoff contracts, technical documentation, peer review

Also: TypeScript, JavaScript, Rust, Next.js; Claude Code (skills, hooks, subagents), Cursor

EXPERIENCE

Full-Stack Developer Intern — Vayaa
Sep 2026 – present

Melbourne, VIC · hybrid · company website: React + TypeScript, Python (FastAPI) + PostgreSQL, deployed on AWS

- Building the company's website from a mentor's brief; turned each plain-language request into numbered functional and non-functional requirements and kept a decision log (choice, alternatives, cost) for every design call.
- Built the contact-enquiry API in Rust (server-side validation, clean types), then led its port to Python (FastAPI) on the mentor's call, gating the cutover on 50 parity tests passing against both backends.
- Took it live on AWS with CloudFormation (CloudFront → ECS Fargate → RDS PostgreSQL) in five attempts, fixing Free-plan limits, a CloudFront account block and RDS TLS (verify-full); SNS emails each enquiry.

Tools: Python, FastAPI, SQLAlchemy, PostgreSQL, Docker, AWS, Rust (Axum), React, TypeScript, Claude Code

AI/ML Engineering Intern — Prodegee
Mar 2026 – Jun 2026

Remote, Melbourne · RMIT industry capstone, Team DS4 — building-footprint inference for off-grid electrification planning

- Built a Claude API model router that classifies task complexity with a cheap Haiku call and routes to Haiku / Sonnet / Opus, keeping cost proportional to difficulty.
- Designed and built the school-geolocation algorithm at the core of the team's pipeline (7.2M Overture building polygons, BFS spatial clustering, composite scoring); ran it on 18,121 schools in Côte d'Ivoire and ported it to Burundi and Chad.
- Built an IoU evaluation harness for compound polygons and traced all three generators' failures to a ~35× over-scoped upstream cluster rather than a fitting fault, which redirected the next assignment.
- Built a ±20% sensitivity harness that identified cluster radius as the dominant parameter (±36–44% swing across 191 villages), driving a 75 → 200 m recalibration that fixed a 90%+ household under-count (0.069 → 0.415).
- Co-built a 40-school satellite-verified ground-truth set and used it to correct a published baseline (6–13% → 37%); published a negative result showing a proximity penalty hurt accuracy at every threshold.
- Trained an XGBoost multi-output tier regressor over 200 villages in 7 Mauritanian Wilayas; documented that its 0.816 "confidence" measured agreement with the training heuristic, not accuracy.
- Replaced a binary distance cutoff with exponential-decay scoring and a weighted confidence score; added per-country sigmoid normalisation so one threshold works in all three countries.
- Led 3 of 10 team assignments and was primary author of 3 of 4 formal deliverables (68 commits, 7 peer reviews); pinned upstream inputs by commit SHA for reproducible re-runs and authored cross-team handoff contracts.

Tools: Python, pandas, scikit-learn, XGBoost, GeoPandas, Kestra, Anthropic API, Git, Claude Code (MCP, hooks)

Dispatch Team Member — Amart Furniture
Feb 2025 – Mar 2026

Melbourne, VIC · part-time, on-site, concurrent with full-time study

- Tracked and reconciled stock in the inventory system, processed customer orders and resolved fulfilment discrepancies across 5 Melbourne stores; trained a new team member.

KEY PROJECTS

Gemma Delegation System — AI agents and fine-tuning · Node.js, MCP SDK, Ollama, Unsloth/QLoRA *May 2026*

- Built an MCP server that lets Claude Code or Cursor delegate lightweight tasks to a locally hosted Gemma model instead of spending frontier-model tokens, failing open to the frontier model whenever Ollama is unavailable.
- Closed the loop: accept/reject feedback updates a capability profile and curates a JSONL dataset; a Colab notebook fine-tunes on confirmed examples (4-bit QLoRA, LoRA r=16) and exports GGUF back into Ollama.
  ↳ GitHub: github.com/faraaz817/gemma-delegation-system

Bro — RAG portfolio chatbot · Next.js, TypeScript, Prisma/Postgres, Anthropic API *2026*

- Built a recruiter-facing site that answers questions from a profile corpus via server-side retrieval with anti-hallucination constraints; deployed on Vercel.
  ↳ Live: job-go-mauve.vercel.app/portfolio

Agent skills for Claude Code / Cursor · reusable accelerators *2026*

- Authored five reusable agent skills others can pick up and run — CO-STAR requirements builder, Socratic tutor, better-way reviewer, skill sync, and a pikepdf CLI that unlocks signature-locked PDFs.
  ↳ GitHub: github.com/faraaz817/My_Skills
- **MGM Prayer Alerts — Android app in community use:** wrote functional requirements before coding, reviewed the code against them and fixed a critical AM/PM storage bug; added GitHub Actions to test, build and publish releases. (Kotlin, JUnit) github.com/faraaz817/MgmPrayerAlerts

EDUCATION

Master of Artificial Intelligence — RMIT University *Jul 2024 – Jul 2026*
Melbourne, VIC · AI Systems Design, Machine Learning, Computer Vision, Intelligent Decision Making (ASP/Clingo), IoT

Bachelor of Technology (Information Technology) — Keshav Memorial Institute of Technology *Dec 2020 – May 2024*
Hyderabad, India (affiliated to JNTUH) · Coursework incl. Machine Learning, Neural Networks & Deep Learning, Data Mining

PUBLICATION

K. M. Asudaria, **M. F. Abdul Khadeer**, S. S. Rafai, K. S. Reddy. "A Comparative Study of Edge Detection Techniques to Identify Maize Leaf Diseases using Machine Learning." *Quest Journal of Software Engineering and Simulation*, Vol. 9(12), pp. 43–51, Dec 2023 (peer-reviewed). Best result Sobel + MLP at 94.75%.

CERTIFICATIONS, AWARDS & LEADERSHIP

- Head, Vachan Speakers Club, KMIT (2022–23) — led public-speaking workshops, debates and communication seminars.
- Smart India Hackathon delegate, Government of India (2022) · Certificate of Publication, Quest Journals (2023).

LANGUAGES

English (native / bilingual) • Urdu (native / bilingual) • Hindi (full professional) • Telugu (elementary)

REFEREES

Available on request.